REVIEW DRAFT — NOT FOR FABRICATION

Six edge-aligned FOC axes / Teensy 4.1 / 12V wrist / two-layer carrier

Global labels join identically named wires across pages. Rectangles show physical connector or component pins.

Power wires approved. Pending: physical module fit, encoder cable verification, component qualification and routing.

01 / Socketed Teensy and signal allocation

File: controller.kicad_sch

02 / Normally-closed hardware kill

File: safety.kicad_sch

03 / Axis 1: index_curl_1

File: axis1.kicad_sch

04 / Axis 2: index_curl_2

File: axis2.kicad_sch

05 / Axis 3: index_splay

File: axis3.kicad_sch

06 / Axis 4: middle_curl_1

File: axis4.kicad_sch

07 / Axis 5: middle_curl_2

File: axis5.kicad_sch

08 / Axis 6: middle_splay

File: axis6.kicad_sch

09 / Power entry and 5V supply

File: power.kicad_sch

10 / STS3215 12V half-duplex interface

File: wrist.kicad_sch

11 / Ground star, expansion and decoupling

File: io.kicad_sch

12 / Test points and release holds

File: test.kicad_sch

13 / Rail and driver-enable indicators

File: indicators.kicad_sch

Sheet: /
File: trueskate-main-v1.kicad_sch

Title: TrueSkate main control PCB

Size: A3	Date:	Rev: v1 DRAFT
KiCad E.D.A. 10.0.6		Id: 1/14

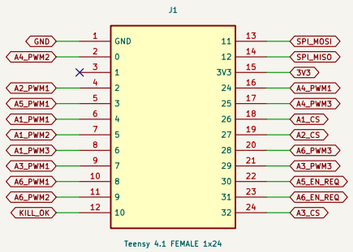
All pin numbers are Teensy numbers. Socket contact names are physical pad IDs.

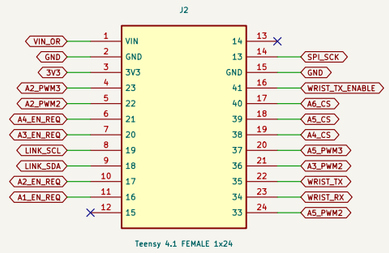
18 PWM outputs use edge alignment at one common frequency: initially 25 kHz.

USB/VIN link MUST be cut on Teensy before simultaneous board and USB power.

J3 is a wire from Teensy VUSB, not a connection to its USB-host 5V pad.

This is a review draft. Existing firmware still uses the old one-axis wiring.







Sheet: /01 / Socketed Teensy and signal allocation/
File: controller.kicad_sch

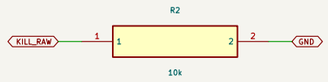
Title: 01 / Socketed Teensy and signal allocation

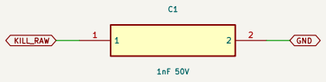
Size: A3	Date:	**Rev: v1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 2/14

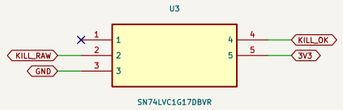
Close NC loop to request permission; open loop/broken cable forces all EN low.
U1/U2 AND each EN request with KILL_OK. Pulldowns keep requests off at boot.
U3 has a Schmitt input: its two switching thresholds reject slow/noisy edges.
NON-LATCHING: restoring the switch can restart a motor if its request is high.
Servo supply is also removed by Q2. This is not a safety-rated stop circuit.



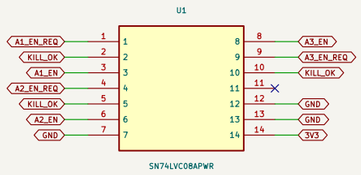


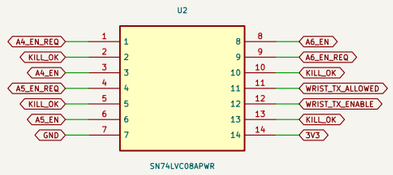


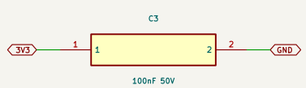


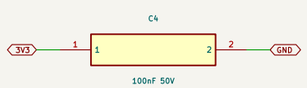


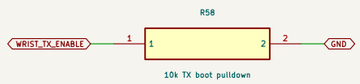












Sheet: /02 / Normally-closed hardware kill/
File: safety.kicad_sch

Title: 02 / Normally-closed hardware kill

Size: A3	Date:	**Rev: v1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 3/14

Female sockets at exact upstream v1.0 coordinates; verify against your physical Mini.

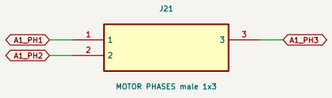
Mini pin 2 is its own regulator OUTPUT. NEVER connect it to Teensy 3.3V.

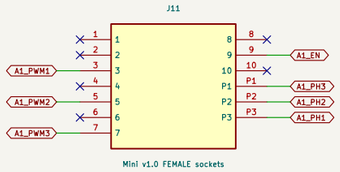
Mini ground pins 1/4 left open: ground reference comes through the paired power wires.

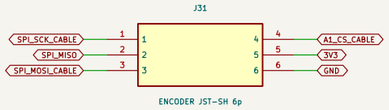
Mini power: approved short screw-terminal wire pair, 12V plus dedicated motor return.

Encoder contact order verified on bench: 1 CLK, 2 MISO, 3 MOSI, 4 CSn, 5 VDD, 6 GND.

Before plugging any encoder in: meter socket contact 5 for 3.3V and contact 6 for ground.





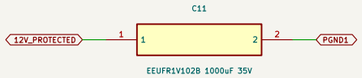


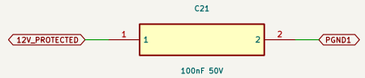














Sheet: /03 / Axis 1: index_curl_1/
File: axis1.kicad_sch

Title: 03 / Axis 1: index_curl_1

Size: A3	Date:	**Rev: v1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 4/14

Female sockets at exact upstream v1.0 coordinates; verify against your physical Mini.

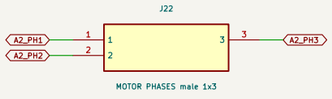
Mini pin 2 is its own regulator OUTPUT. NEVER connect it to Teensy 3.3V.

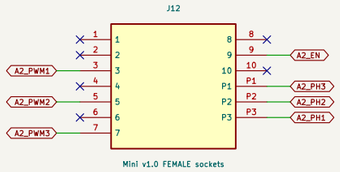
Mini ground pins 1/4 left open: ground reference comes through the paired power wires.

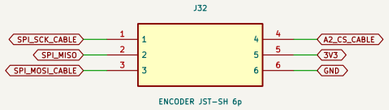
Mini power: approved short screw-terminal wire pair, 12V plus dedicated motor return.

Encoder contact order verified on bench: 1 CLK, 2 MISO, 3 MOSI, 4 CSn, 5 VDD, 6 GND.

Before plugging any encoder in: meter socket contact 5 for 3.3V and contact 6 for ground.





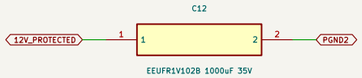


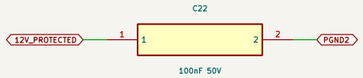














Sheet: /04 / Axis 2: index_curl_2/
File: axis2.kicad_sch

Title: 04 / Axis 2: index_curl_2

Size: A3	Date:	**Rev: v1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 5/14

Female sockets at exact upstream v1.0 coordinates; verify against your physical Mini.

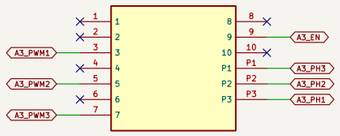
Mini pin 2 is its own regulator OUTPUT. NEVER connect it to Teensy 3.3V.

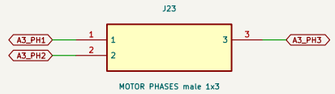
Mini ground pins 1/4 left open: ground reference comes through the paired power wires.

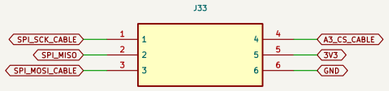
Mini power: approved short screw-terminal wire pair, 12V plus dedicated motor return.

Encoder contact order verified on bench: 1 CLK, 2 MISO, 3 MOSI, 4 CSn, 5 VDD, 6 GND.

Before plugging any encoder in: meter socket contact 5 for 3.3V and contact 6 for ground.





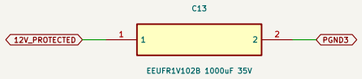


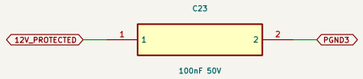














Sheet: /05 / Axis 3: index_splay/
File: axis3.kicad_sch

Title: 05 / Axis 3: index_splay

Size: A3	Date:	**Rev: v1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 6/14

Female sockets at exact upstream v1.0 coordinates; verify against your physical Mini.

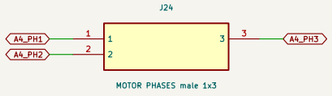
Mini pin 2 is its own regulator OUTPUT. NEVER connect it to Teensy 3.3V.

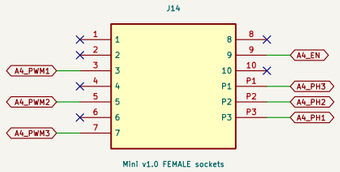
Mini ground pins 1/4 left open: ground reference comes through the paired power wires.

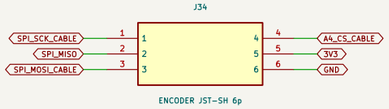
Mini power: approved short screw-terminal wire pair, 12V plus dedicated motor return.

Encoder contact order verified on bench: 1 CLK, 2 MISO, 3 MOSI, 4 CSn, 5 VDD, 6 GND.

Before plugging any encoder in: meter socket contact 5 for 3.3V and contact 6 for ground.





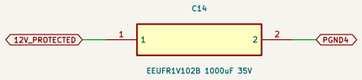


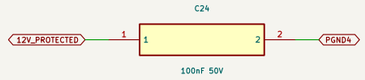














Sheet: /06 / Axis 4: middle_curl_1/
File: axis4.kicad_sch

Title: 06 / Axis 4: middle_curl_1

Size: A3 | Date: | **Rev: v1 DRAFT**

KiCad E.D.A. 10.0.6 | Id: 7/14

Female sockets at exact upstream v1.0 coordinates; verify against your physical Mini.

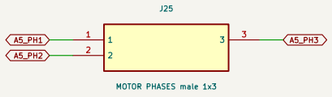
Mini pin 2 is its own regulator OUTPUT. NEVER connect it to Teensy 3.3V.

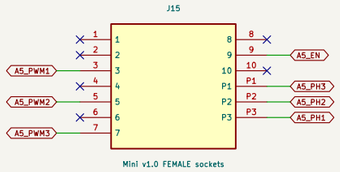
Mini ground pins 1/4 left open: ground reference comes through the paired power wires.

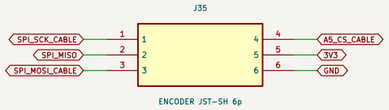
Mini power: approved short screw-terminal wire pair, 12V plus dedicated motor return.

Encoder contact order verified on bench: 1 CLK, 2 MISO, 3 MOSI, 4 CSn, 5 VDD, 6 GND.

Before plugging any encoder in: meter socket contact 5 for 3.3V and contact 6 for ground.







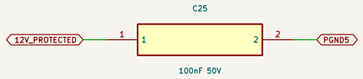














Sheet: /07 / Axis 5: middle_curl_2/
File: axis5.kicad_sch

Title: 07 / Axis 5: middle_curl_2

Size: A3	Date:	**Rev: v1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 8/14

Female sockets at exact upstream v1.0 coordinates; verify against your physical Mini.

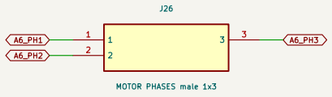
Mini pin 2 is its own regulator OUTPUT. NEVER connect it to Teensy 3.3V.

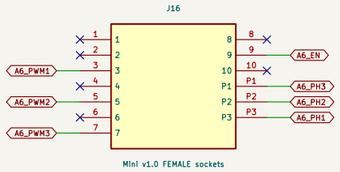
Mini ground pins 1/4 left open: ground reference comes through the paired power wires.

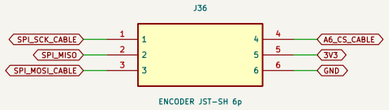
Mini power: approved short screw-terminal wire pair, 12V plus dedicated motor return.

Encoder contact order verified on bench: 1 CLK, 2 MISO, 3 MOSI, 4 CSn, 5 VDD, 6 GND.

Before plugging any encoder in: meter socket contact 5 for 3.3V and contact 6 for ground.





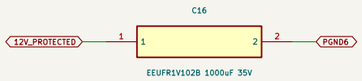


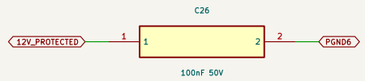














Sheet: /08 / Axis 6: middle_splay/
File: axis6.kicad_sch

Title: 08 / Axis 6: middle_splay

Size: A3	Date:	Rev: v1 DRAFT
KiCad E.D.A. 10.0.6		Id: 9/14

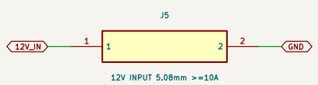
Input: regulated 12V ONLY. Initial design envelope 5A total, validate fuse/inrush/temperature.

Q1 is reversed relative to a normal switch: body diode starts forward input, blocks reversal.

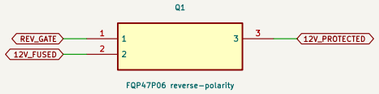
D1/D2 OR the two 5V sources. CUT Teensy VUSB–VIN link or the PC can be back–powered.

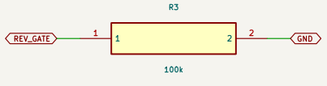
Buck layout needs a very short VIN/CIN/GND loop and a compact SW/inductor connection.

Input and local buck/axis copper are hand–routed; main distribution remains unfinished.

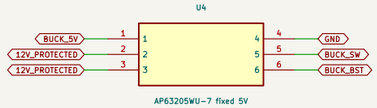




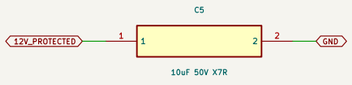


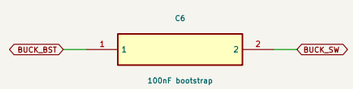




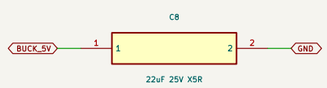


























Flags declare power through passive components.

They do not add physical parts or prove safe wiring.

Sources: PSU, Teensy regulator, protected 12V,

diode–selected VIN and regulated buck 5V.

Sheet: /09 / Power entry and 5V supply/
File: power.kicad_sch

Title: 09 / Power entry and 5V supply

Size: A3	Date:	**Rev: v1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 10/14

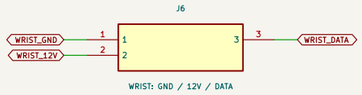
Serial8: RX34, TX35. Pin41 HIGH enables transmit; LOW releases shared data wire.

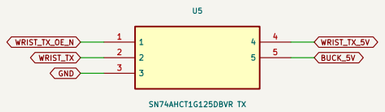
5V AHCT transmit buffer accepts 3.3V logic. LVC receiver accepts 5V at its input.

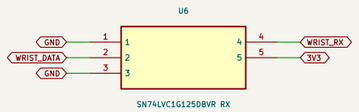
Q3 defaults transmit OFF; U2 also forces transmit OFF during kill. R8 is DNP.

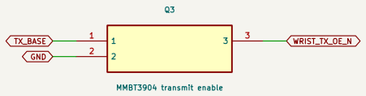
Feetech connector footprint is PROVISIONAL PH 2.0mm; verify your actual cable.

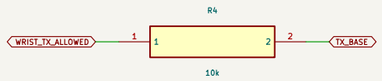
Q2 cuts servo 12V when KILL_OK is low. Logic ground stays connected.

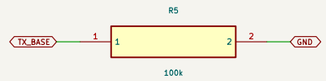


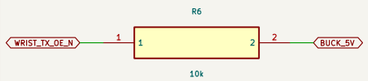






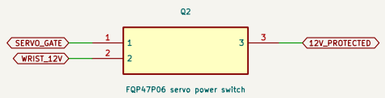


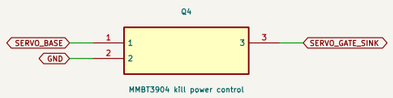






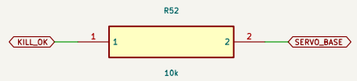


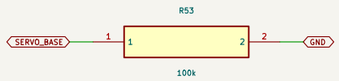
















Sheet: /10 / STS3215 12V half-duplex interface/
File: wrist.kicad_sch

Title: 10 / STS3215 12V half-duplex interface

Size: A3	Date:	**Rev: v1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 11/14

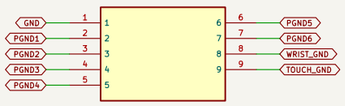
NT1 is a copper net-tie: all grounds meet only at this planned power-entry star.

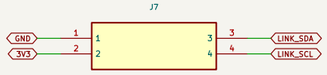
PGND1..6 are separate motor return branches. Do not join through Mini control sockets.

Touch returns have their own branch plus one ferrite bead each, away from motor current.

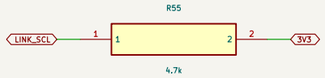
I2C expansion is 3.3V only; 4.7k pullups. No daughterboard motor power on this connector.

Star is beside input; buck return is routed. Motor, wrist and touch branches remain open.











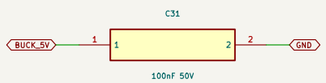


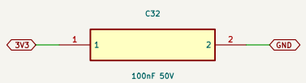






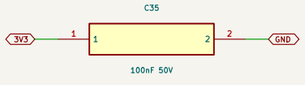




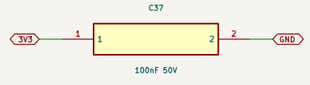


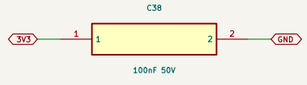












Sheet: /11 / Ground star, expansion and decoupling/
File: io.kicad_sch

Title: 11 / Ground star, expansion and decoupling

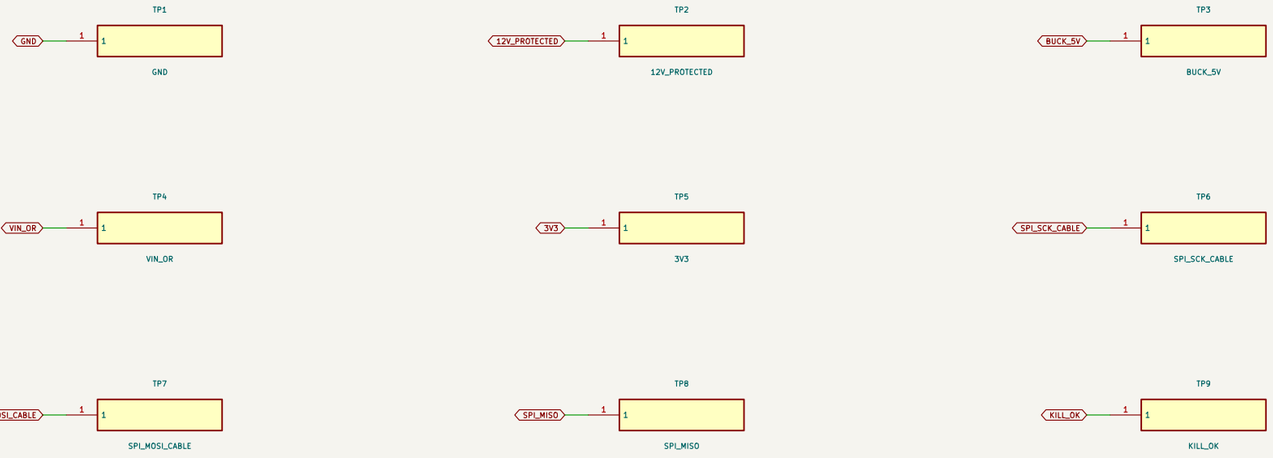
Size: A3	Date:	**Rev: v1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 12/14

REVIEW DRAFT: physical module fit and encoder cable pinout unresolved.

Partial power routing and local buck ground only; no assembly file or Gerbers released.

Do not treat ERC success as proof of correct connectors, thermal design or firmware.

See docs/pcb/design-notes.md and verification.md before considering manufacture.



Sheet: /12 / Test points and release holds/
File: test.kicad_sch

Title: 12 / Test points and release holds

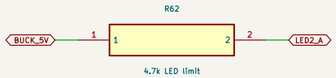
Size: A3	Date:	Rev: v1 DRAFT
KiCad E.D.A. 10.0.6		Id: 13/14

Red LEDs show rail presence and hardware-gated enables, not actual motor motion.

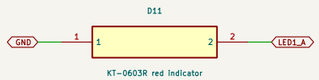
Series resistors limit LED current. Enable LEDs use small current to avoid loading EN.

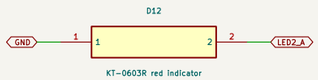
An extinguished LED is NOT proof of a discharged or safe circuit; use a meter.

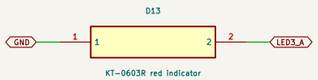


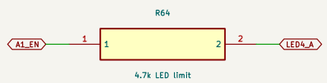


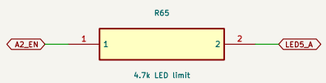


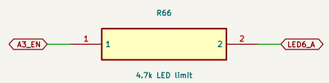


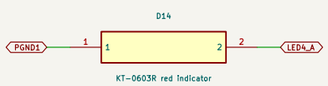


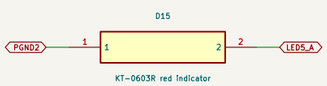


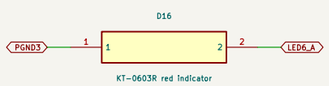




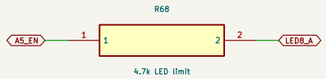




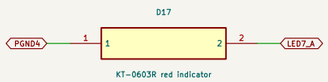




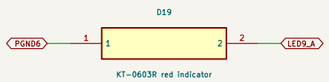












Sheet: /13 / Rail and driver-enable indicators/
File: indicators.kicad_sch

Title: 13 / Rail and driver-enable indicators

Size: A3	Date:	**Rev: v1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 14/14